# BioTraj
# User Manual
# Version 1.2

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# 1 Introduction

BioTraj (BioTrajectoryAnalysis) is an open-source collection of Python tools for extracting and quantitatively analysing biological trajectories from video recordings. The framework combines computer vision, automated tracking, conventional trajectory descriptors, and information-geometric analysis of spatially and temporally resolved movement distributions.

The software was initially developed for ant trajectory analysis, but the workflow can be adapted to other organisms and moving agents provided that suitable two-dimensional trajectories can be obtained. Tracking and analysis are separated through CSV trajectory files, so trajectories generated by other tracking programs can also be analysed.

The current release contains two principal components:

1. `AntTracking_V1.0.py`: video-based trajectory extraction;
2. `AntTrajectoryAnalysis_V1.2_info_geometry_final.py`: conventional trajectory analysis together with spatial and temporal Fisher–Rao information geometry.

Companion utilities are used for scene annotation and annotation-aware interpretation when required.

# 2 Software Architecture

The BioTraj workflow is modular: video acquisition and tracking produce trajectory coordinates, while the analysis program operates on exported trajectory files. This separation permits independent use of the analysis module and makes it possible to add new descriptors without changing the tracking program.

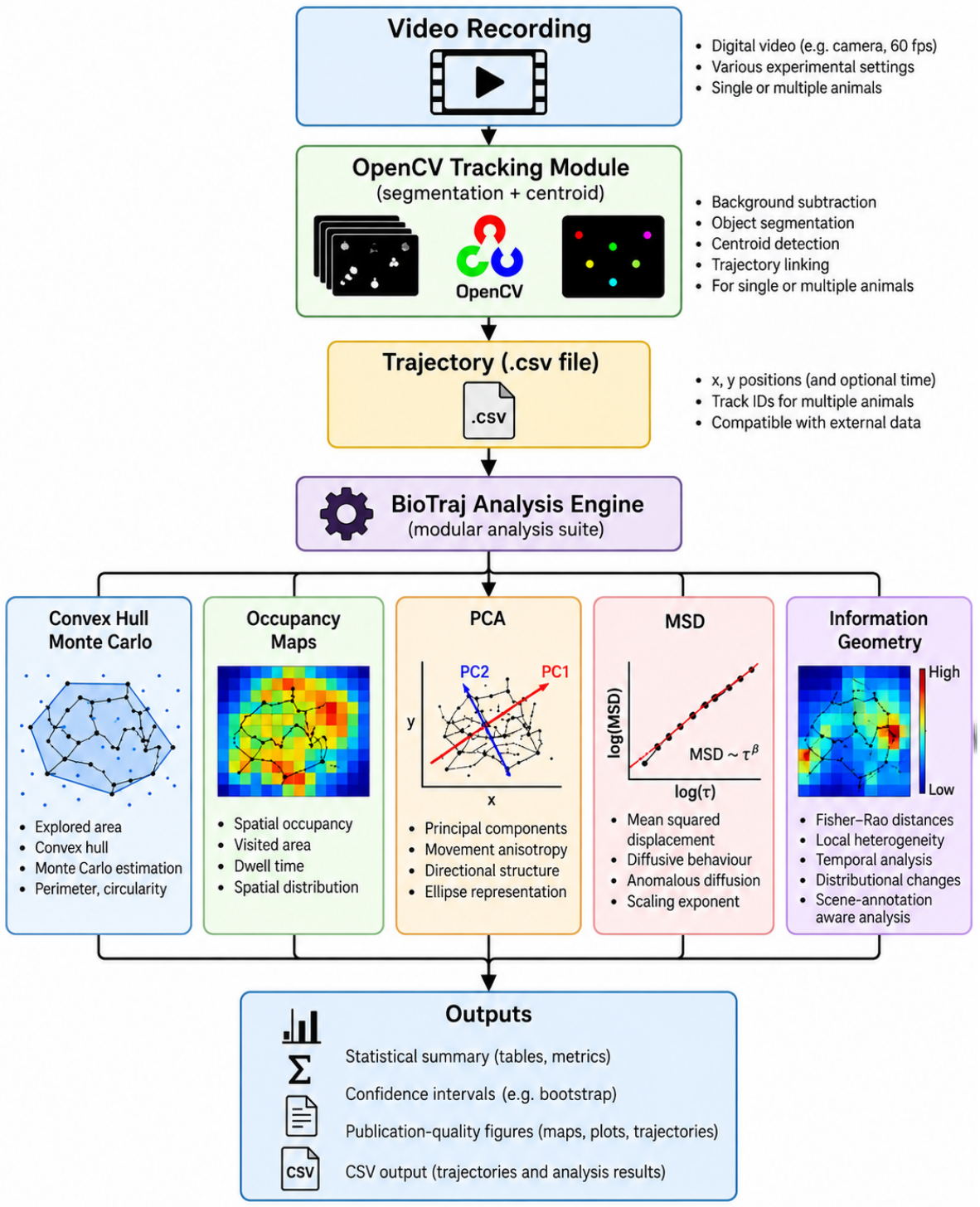



Figure 1: Overview of the BioTraj workflow. Video-derived trajectories are exported to CSV and analysed using conventional geometric and dynamical descriptors together with spatial and temporal information geometry.

## 3 Installation

BioTraj requires Python 3.10 or newer. Principal dependencies include OpenCV, NumPy, Pandas, SciPy, Matplotlib, and scikit-learn. Install repository requirements using:

```
pip install -r requirements.txt
```

# 4 Coordinate System

Video images follow the OpenCV convention, with the origin at the upper-left corner. Statistical analyses and graphical outputs may transform coordinates to a Cartesian representation with the origin at the lower-left, the $x$ axis increasing to the right, and the $y$ axis increasing upwards. Users should retain the original image coordinate system when defining scene annotations intended to overlay the video frame.

# 5 Trajectory Extraction

## 5.1 Overview

The tracking program detects moving objects, filters detections, associates detections between frames, reconstructs continuous track segments, and writes coordinates to a CSV file. A tracking identifier denotes a continuous reconstructed track segment; it should not automatically be interpreted as a persistent biological individual when identities can be lost or exchanged.

## 5.2 Basic Usage

```
python3 AntTracking_V1.0.py video.mov
```

A typical configured run is:

```
python3 AntTracking_V1.0.py video.mov \
    --threshold 18 \
    --min-area 6 \
    --max-area 250 \
    --output tracking_results
```

Important tracking parameters include the detection threshold, minimum and maximum object area, maximum association distance, maximum permitted track gap, and merge distance. Appropriate values depend on image scale, contrast, frame rate, and organism size.

# 6 Trajectory Data Format

The analysis program accepts trajectory data in CSV format. A typical file contains:

```
track_id,frame,x,y
1,0,123.4,201.7
1,1,124.2,202.3
1,2,125.0,203.1
```

Coordinates may be expressed in pixels or calibrated physical units. Turning angles are calculated only within uninterrupted track segments and are never bridged across missing portions of a trajectory.

# 7 Trajectory Analysis

## 7.1 Basic Usage

The current analysis program is run with the trajectory CSV as the positional argument:

```
python3 AntTrajectoryAnalysis_V1.2_info_geometry_final.py \
    ant_tracks.csv \
    --output analysis_results
```

Without information-geometry switches, the program provides the conventional BioTraj trajectory descriptors and associated figures.

## 7.2 Conventional Descriptors

BioTraj calculates complementary geometric and dynamical descriptors, including convex-hull explored area, Monte Carlo area estimation, occupancy maps, principal-component anisotropy, mean-squared displacement (MSD), straightness, and fractal dimension.

### 7.2.1 Explored Area and Monte Carlo Estimation

The explored domain is approximated by the convex hull. A Monte Carlo estimate can be obtained from random points within a bounding rectangle:

$$A_{\mathrm{MC}} = A_{\mathrm{box}} \frac{N_{\mathrm{inside}}}{N_{\mathrm{total}}}. \tag{1}$$

### 7.2.2 Principal Component Analysis

For trajectory positions $\mathbf{r}_i$, the covariance matrix is diagonalized to obtain eigenvalues $\lambda_1$ and $\lambda_2$. The anisotropy ratio is

$$A = \frac{\lambda_1}{\lambda_2}. \tag{2}$$

Values near unity indicate approximately isotropic spatial exploration; larger values indicate an increasingly elongated spatial distribution.

### 7.2.3 Mean-Squared Displacement

$$\mathrm{MSD}(\tau) = \left\langle |\mathbf{r}(t+\tau) - \mathbf{r}(t)|^2 \right\rangle. \tag{3}$$

When an approximate power law $\mathrm{MSD}(\tau) \propto \tau^{\beta}$ is observed, $\beta < 1$ corresponds to sublinear/-subdiffusive scaling, $\beta \approx 1$ to diffusive scaling, and $\beta > 1$ to superlinear/superdiffusive scaling. Superlinear scaling can arise from persistent or otherwise correlated motion and should not be assigned a unique mechanism from $\beta$ alone.

### 7.2.4 Straightness and Fractal Dimension

Straightness compares net displacement with total path length. Fractal dimension provides a complementary measure of trajectory complexity and space-filling character over the analysed scales.

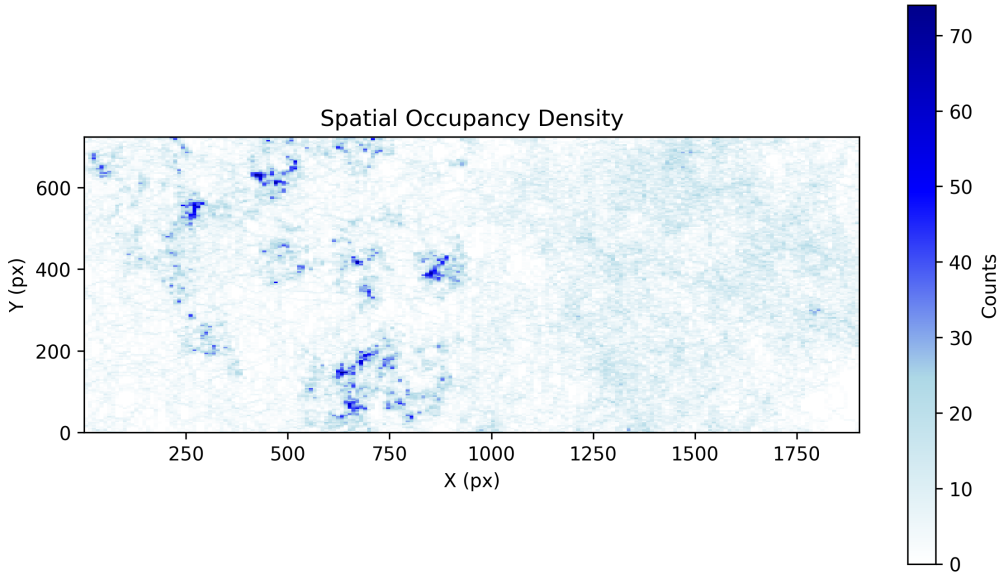



Figure 2: Example BioTraj occupancy map. Occupancy describes *where* observations occur and is complementary to the information-geometric analysis, which describes how local movement distributions differ.

# 8 Information-Geometric Analysis

## 8.1 Purpose and Interpretation

The information-geometric module detects spatial or temporal changes in empirical movement distributions without requiring predefined behavioural states. The present implementation uses unsigned turning-angle distributions.

**Important interpretation rule:** Fisher–Rao heterogeneity identifies where or when estimated movement statistics change; it does not determine why they change. Elevated values may reflect biological responses, physical constraints, surface geometry, projection effects, tracking artefacts, finite sampling, or combinations of these factors.

## 8.2 Turning-Angle Distributions

For three consecutive positions in the same continuous track segment,

$$\Delta \mathbf{r}_t = \mathbf{r}_{t+1} - \mathbf{r}_t, \tag{4}$$

and the unsigned turning angle is

$$\Delta\theta_t = \arccos\left(\frac{\Delta \mathbf{r}_{t-1} \cdot \Delta \mathbf{r}_t}{\|\Delta \mathbf{r}_{t-1}\| \, \|\Delta \mathbf{r}_t\|}\right), \qquad 0 \le \Delta\theta \le \pi. \tag{5}$$

Within a spatial cell or temporal window, the angles are converted to a normalized histogram on a fixed common angular partition,

$$P = (p_1, \ldots, p_K), \qquad p_k \ge 0, \qquad \sum_{k=1}^{K} p_k = 1. \tag{6}$$

## 8.3 Fisher–Rao Distance

The square-root representation

$$\psi(P) = (\sqrt{p_1}, \ldots, \sqrt{p_K}) \tag{7}$$

maps the categorical probability simplex to the positive orthant of the unit sphere. Under the metric convention used by BioTraj, the Fisher–Rao geodesic distance between two categorical distributions is

$$d_{\mathrm{FR}}(P, Q) = 2 \arccos\left(\sum_{k=1}^{K} \sqrt{p_k q_k}\right). \tag{8}$$

The same fixed histogram partition must be used when distributions are compared.

## 8.4 Spatial Information Geometry

With `-info-geometry`, the analysed domain is divided into a rectangular grid. A turning-angle distribution is estimated for each sufficiently sampled cell. Pairwise Fisher–Rao distances quantify statistical separation between cells, while the local Fisher–Rao heterogeneity assigned to a cell summarizes its distances from valid neighbouring cells.

Large local values indicate that a cell's estimated turning-angle distribution differs strongly from those of its neighbours. Small values indicate locally similar distributions.

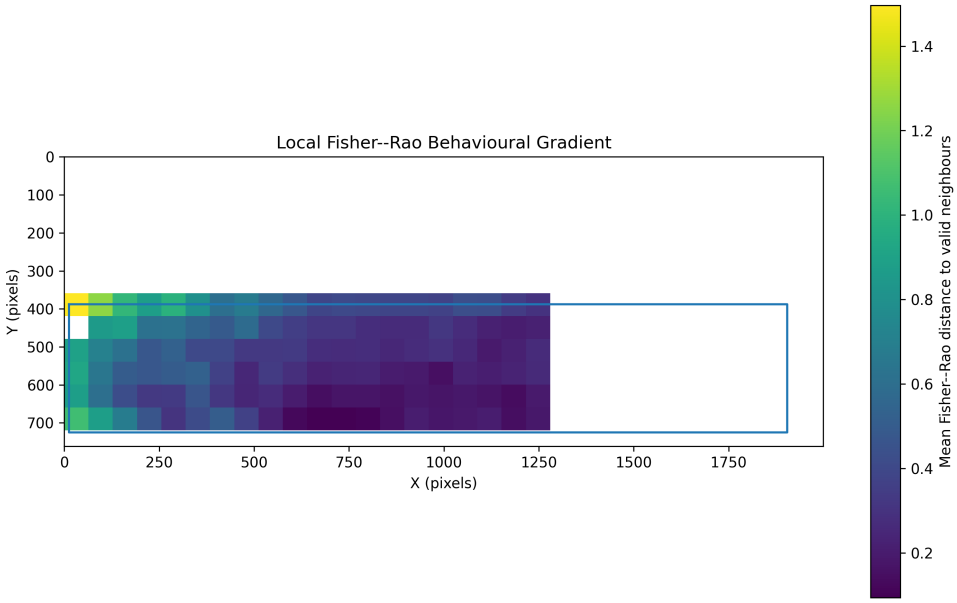



Figure 3: Example spatial Fisher–Rao map. Colour represents local Fisher–Rao heterogeneity calculated from neighbouring spatial cells. The map identifies spatial changes in movement statistics and should not be interpreted automatically as a behavioural-state map.

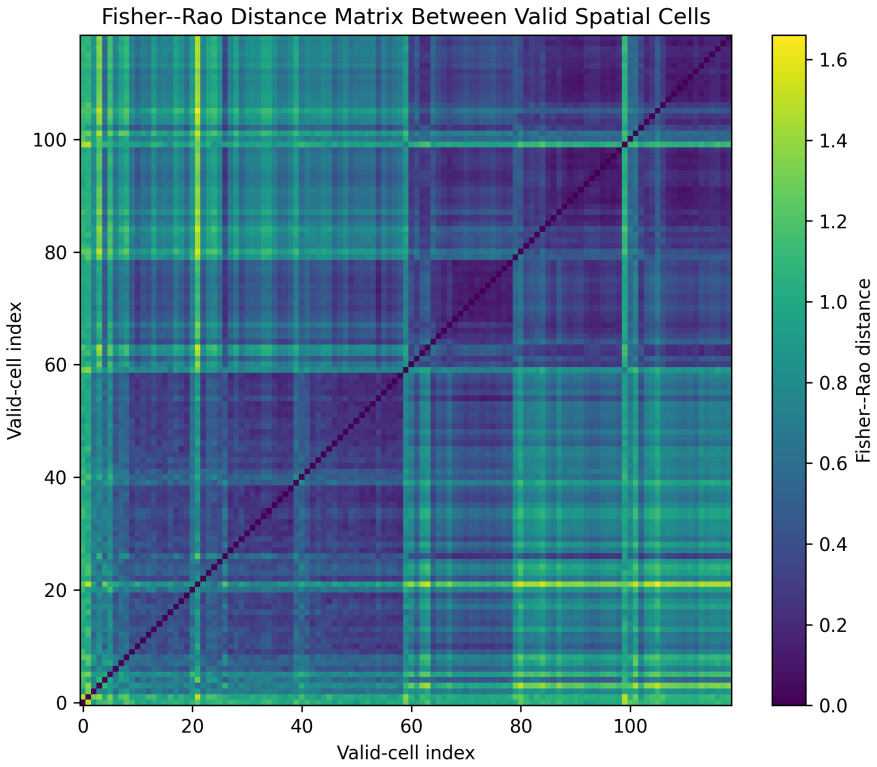



Figure 4: Example pairwise Fisher–Rao distance matrix between sufficiently sampled spatial cells. The matrix represents statistical similarity independently of physical separation; the local spatial map in Figure 3 restores neighbourhood context.

### 8.5 Spatial Sampling Diagnostics

Cells containing fewer than `-info-min-samples` valid turning angles are excluded. The sampling map should always be inspected together with the Fisher–Rao map because empirical histogram distances become more variable when estimated from few observations.

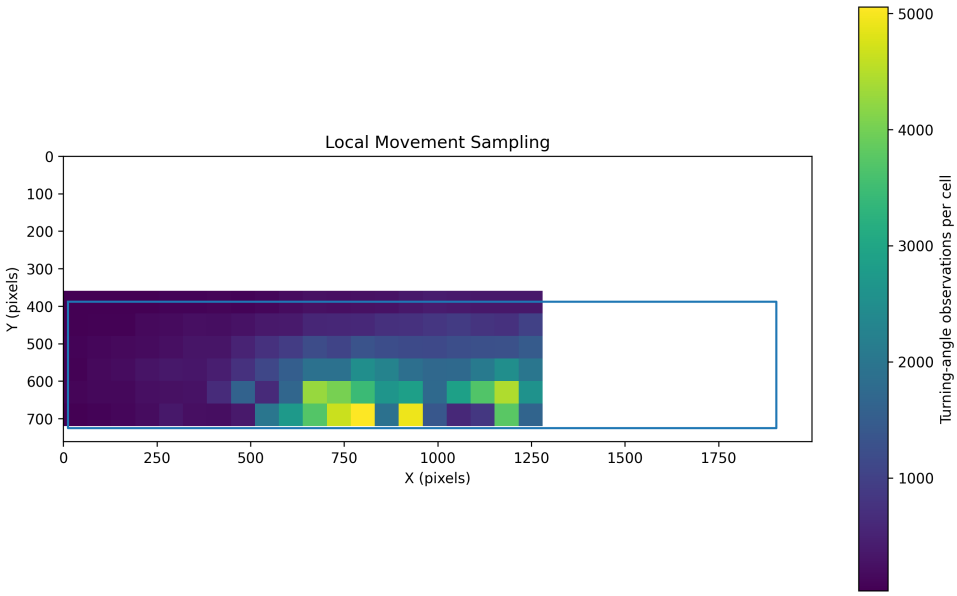



Figure 5: Example spatial sampling map showing the number of valid turning-angle observations available in each grid cell. Cells below the configured minimum sample count are excluded from information-geometric comparisons.

## 8.6 Temporal Information Geometry

With `-temporal-info-geometry`, the recording is divided into temporal windows. A turning-angle distribution is estimated for each sufficiently sampled window. BioTraj writes the complete temporal Fisher–Rao distance matrix and the distances between consecutive windows.

The window length controls temporal resolution. A sharp transition occurring inside a window produces a mixed distribution and may therefore yield the largest consecutive-window distance in the following comparison rather than exactly at the physical transition time.

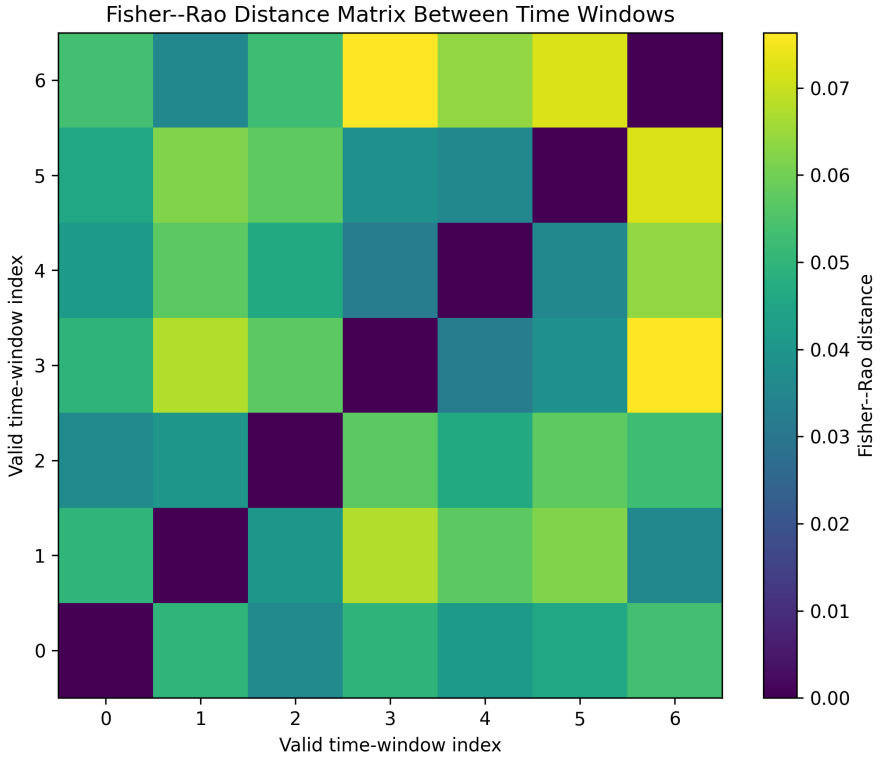



Figure 6: Example temporal Fisher–Rao distance matrix. Each element compares the turning-angle distributions estimated in two temporal windows.

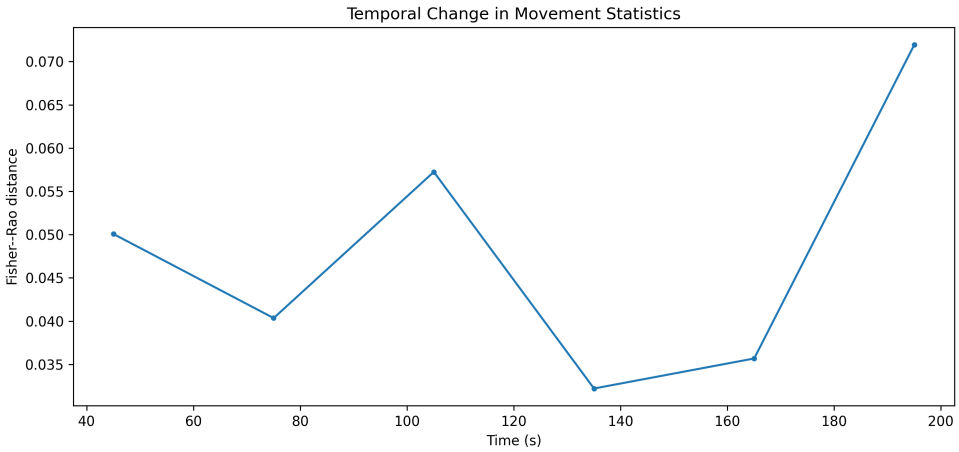



Figure 7: Example consecutive-window Fisher–Rao analysis. Peaks identify periods in which the estimated movement distribution changes strongly between adjacent temporal windows.

### 8.7 Bootstrap and Finite-Sampling Controls

The spatial and temporal modules can bootstrap empirical turning-angle samples to estimate uncertainty in Fisher–Rao distances. This quantifies empirical sampling variability but does not make temporally correlated observations or neighbouring spatial cells independent biological replicates.

Raw Fisher–Rao values should not be used to rank datasets with very different local sample sizes without additional sampling controls. Equal-count resampling and threshold-convergence analyses are useful diagnostics.

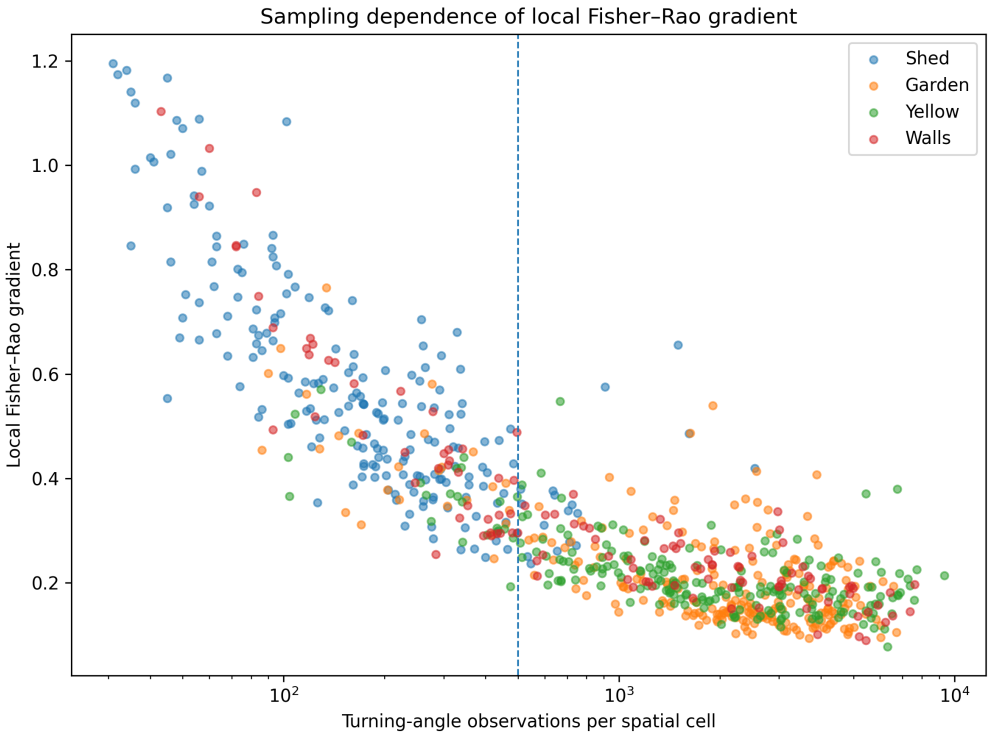



Figure 8: Illustration of finite-sampling effects on local Fisher–Rao heterogeneity. Sparse cells can show larger and more variable empirical distances; sampling diagnostics are therefore part of the recommended interpretation workflow.

## 9 Information-Geometry Command-Line Options

The principal options in BioTraj v1.2 are summarized below.

| Option | Default | Meaning |
|---|---|---|
| `-info-geometry` | off | Enable spatial information geometry. |
| `-info-grid-x` | 20 | Number of spatial grid columns. |
| `-info-grid-y` | 12 | Number of spatial grid rows. |
| `-info-angle-bins` | 18 | Number of bins over 0–180°. |
| `-info-min-samples` | 30 | Minimum turning-angle observations per spatial cell. |
| `-info-clusters` | 4 | Number of hierarchical information-geometric distributional clusters. Clustering is optional/descriptive and does not define biological states. |
| `-info-pseudocount` | 0 | Optional pseudocount added per angular bin. |

| Option | Default | Meaning |
|---|---|---|
| `-info-bootstrap` | 200 | Spatial bootstrap replicates; 0 disables bootstrap. |
| `-info-seed` | 12345 | Random seed for information-geometry bootstrap. |
| `-temporal-info-geometry` | off | Enable temporal information geometry. |
| `-temporal-window-frames` | 1800 | Temporal window length in frames. |
| `-temporal-step-frames` | window length | Step between temporal windows. |
| `-temporal-min-samples` | 100 | Minimum turning-angle observations per temporal window. |
| `-temporal-bootstrap` | 200 | Temporal bootstrap replicates; 0 disables bootstrap. |
| `-fps` | none | Frame rate; when supplied, temporal axes can be expressed in seconds. |

# 10 Reproducing the Information-Geometry Settings Used in the BioTraj Study

The analyses reported with BioTraj v1.2 used a $20 \times 12$ spatial grid, 18 turning-angle bins, a minimum of 30 angles per spatial cell, 200 bootstrap replicates, and 15-s non-overlapping temporal windows for 60-fps recordings. Because the program default temporal window is 1800 frames, the 900-frame manuscript setting must be supplied explicitly:

```
python3 AntTrajectoryAnalysis_V1.2_info_geometry_final.py \
    ant_tracks.csv \
    --output BioTraj_V1_2_results \
    --info-geometry \
    --info-grid-x 20 \
    --info-grid-y 12 \
    --info-angle-bins 18 \
    --info-min-samples 30 \
    --info-clusters 4 \
    --info-pseudocount 0 \
    --info-bootstrap 200 \
    --info-seed 12345 \
    --temporal-info-geometry \
    --temporal-window-frames 900 \
    --temporal-step-frames 900 \
    --temporal-min-samples 100 \
    --temporal-bootstrap 200 \
    --fps 60
```

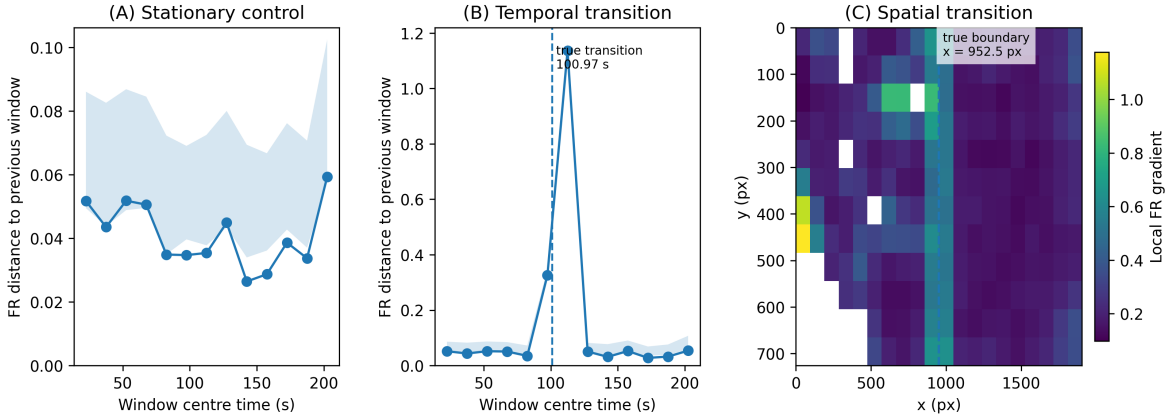



Figure 9: Simulation-based validation example used during development of the information-geometric module. Stationary, temporal-transition, and spatial-transition cases illustrate how known imposed statistical changes are recovered within the resolution of the selected temporal windows and spatial grid.

## 11 Information-Geometry Output Files

Depending on enabled options, the v1.2 analysis writes numerical tables, summaries, and figures. Important information-geometric outputs include:

- `info_geometry_cells.csv`: cell-level spatial information-geometry results;
- `info_geometry_summary.json`: spatial analysis settings and summary quantities;
- spatial sampling, local heterogeneity, distance-matrix, and distributional-cluster figures;
- `temporal_info_windows.csv`: window-level temporal information;
- `temporal_fisher_rao_distance_matrix.csv`: complete temporal distance matrix;
- `temporal_info_geometry_summary.json`: temporal settings and summary information;
- temporal sampling, distance-matrix, and consecutive-distance figures.

File names can evolve slightly between development builds; the release version and its generated summary files should be retained with published analyses.

## 12 Scene Annotation and Annotation-Aware Analysis

Scene annotation is handled by companion BioTraj utilities rather than by the main v1.2 trajectory-analysis command described above. Annotations are defined in the original image coordinate system and stored separately from the trajectory data. They can represent polygonal surfaces or biological regions, polylines marking physical transitions, circular features, observation boundaries, and independently identified artifact regions.

Three concepts should remain distinct:

$$\textbf{tracking domain} \neq \textbf{biological/physical regions} \neq \textbf{artifact masks}.$$

Whenever possible, scene annotations intended for interpretation should be defined independently of the Fisher–Rao result. Annotation-aware comparisons are descriptive unless the experimental design provides genuine independent biological replication.

## 13 Recommended Analysis Workflow

A robust information-geometric workflow is:

1. inspect tracking quality and preserve continuous track segments;
2. inspect occupancy and conventional trajectory descriptors;
3. inspect the spatial sampling map;
4. calculate spatial and/or temporal Fisher–Rao results using justified minimum sample counts;
5. inspect bootstrap uncertainty and finite-sampling diagnostics;
6. relate detected statistical structure to independently defined scene information where appropriate;
7. avoid assigning biological mechanisms or behavioural states from Fisher–Rao values alone.

## 14 Limitations

The current implementation primarily assumes two-dimensional motion in the imaging plane. Tracking can be affected by poor contrast, object overlap, illumination changes, out-of-plane motion, and projection effects. Information-geometric estimates additionally depend on histogram resolution, spatial/temporal partitioning, and finite sample size. Spatial cells and temporal observations from a single recording are generally correlated and should not automatically be treated as independent biological replicates.

## 15 Applications

BioTraj can be adapted to ants and other social insects, protozoa, rotifers, nematodes, slugs and snails, small aquatic organisms, active-matter experiments, and simulated or robotic agents. Suitability depends on whether reliable trajectories can be reconstructed at a spatial and temporal resolution appropriate to the biological question.

## 16 Citation and Reproducibility

If BioTraj is used in academic work, cite the associated publication and the archived/released version of the BioTrajectoryAnalysis repository. For reproducibility, record the software release or commit, command-line options, random seed, input trajectory file, frame rate, calibration, and any scene-annotation files used in interpretation.

## 17 License

BioTraj is distributed under the MIT License.